

The Value of Information in Longevity Biomarkers

Abstract

Noninvasive biomarkers for use in general longevity — i.e., not for the diagnosis of a disease — are increasingly common. Consumer interest in biomarkers has been piqued, and significant funding has entered gerontology, marking a departure from typical investments that historically followed specific infirmities.

Introduction

Biomarkers can provide useful health information to the user. A significant consumer market around biomarkers has emerged, including labs and at-home test kits, which either provide a result to the user directly or ship a sample to a lab for analysis. This market has seen significant growth over the past decade.

Tests can be differentiated by the time horizon of their results, from acute to long-term. There is also a broader shift away from biomarkers used strictly for the diagnosis and treatment of disease, toward optimization in the absence of clinical illness. Biomarkers are increasingly accompanied by accessible information on interventions — typically dietary, lifestyle, and supplement-based — that can act on the biomarker of interest, often outside of medical supervision.

As with most commercial adoption, the use of biomarkers was initially spearheaded by early adopters — "biohackers" and others with a supernormal interest in physiology, such as bodybuilders, athletes, or people seeking to improve cognitive performance. Given this interest, significant self-experimentation, occasionally documented, published, and discussed in forums, has taken place.

Tests such as testosterone have become far more widespread, owing to elevated interest in the health implications of endocrine regulation and awareness of environmental influences (1). Testosterone tests, previously difficult to obtain, have become a mainstay of many integrative biomarker panels. While the accuracy of any given test and the efficacy of proposed interventions may vary, the general approach is sound: measurement, typically at an early stage before disease onset, followed by efforts to optimize the value, and optionally a retest after one or several interventions.

Much of this guidance takes place outside of the medical system, either through the parallel health-coaching ecosystem or performed informally through self-tracking. This "hobbyist" orientation may skirt medical regulation. Individuals interested in biohacking vary significantly in their appetite for risk, and actual risk can vary significantly between well-tested interventions (e.g., drink more water) and untested black- or gray-market peptides.

Biomarkers also vary significantly in accessibility, cost, and feasible testing frequency. Innovations such as the continuous glucose monitor took blood glucose measurement from a discrete event — often requiring travel to a hospital — to a handheld device capable of discrete measurements, to continuous real-time measurement. Wearables similarly measure parameters like heart rate (HR) and heart rate variability (HRV) on a continuous basis, automatically logged as long as the user wears the device, which many do nearly 24/7.

These combined measurements provide rich health data that can be used for optimization. While the cost of many tests has decreased, it remains nonzero. This analysis provides a value-of-information (VOI) framework for assessing the return on investment of a given biomarker test, in order to identify high-value biomarker measurements for the general population and for specific cohorts. This health-economics analysis helps assess whether, and under what conditions, biomarker measurements should be incorporated into health plans, whether public or private.

Biomarkers and Mortality Associations

Across medicine, a wide variety of measurements provide clinical value depending on circumstance. Some are specific to a particular disease and may not generalize to longevity. Where a biomarker is generally associated with longevity, it may be a useful target for the general healthy population to optimize. Existing work has characterized all-cause-mortality (ACM)-associated biomarkers. ACM is a useful summary measure of general health, since most people prefer to live longer rather than shorter, and lifespan is associated with good health during one's life — a concept typically referred to as healthspan, the duration of disease-free life.

Most health-economics framings implicitly adopt this view through use of the disability- or quality-adjusted life year (DALY/QALY) as the quantity optimized by an intervention or policy. In public health, even a modest improvement in individual health outcomes — engendered by, say, anti-smoking laws, walkable public spaces, or breastfeeding awareness — can have a significant impact when applied at the population level.

Methods

Biomarkers

We extract the mortality-associated biomarkers cataloged in the database mortalitypredictors.org and adapt them into a common schema. Where possible, each is linked to a LOINC test ID. This work generalizes previous work (2,3). Price data are extracted from findalabtest.com.

We extract the 471 biomarkers listed on mortalitypredictors.org and retain those for which either (a) a published mapping between biomarker value and mortality hazard ratio (or a suitable proxy) exists, or (b) such a mapping can be calculated from a public dataset (e.g.,

NHANES linked mortality files). Where sufficient data exist, we calculate the population distribution and mortality hazard-ratio function for available age and sex cohorts.

Formal framework

This section formalizes the technique summarized above as a sequence of linked models: (i) a population distribution of the biomarker, (ii) a hazard-ratio function mapping biomarker value to relative mortality risk, (iii) propagation of that hazard ratio into expected years of life lost and DALYs, (iv) a decision-theoretic value-of-information (VOI) layer, and (v) an Expected Hazard Information Value (EHIV) that nets VOI against test cost. Table notation is summarized at the end of this section.

1. Population distribution of the biomarker. For biomarker X , age a , sex s , and (optionally) an additional cohort stratifier c (ethnicity, underlying condition, genotype), we characterize the empirical density

$$(1) \quad X \mid (a, s, c) \sim f_{X}(x; a, s, c)$$

estimated from NHANES or an equivalent population survey, from which an individual's percentile position and z-score relative to their cohort can be recovered.

2. Hazard-ratio function. Published Cox proportional-hazards models (or extractions from mortalitypredictors.org) supply a relative hazard as a function of biomarker value, referenced to a reference value x_{ref} (e.g., cohort median or lowest-risk quantile):

$$(2) \quad \text{HR}(x) = h(x) / h(x_{\text{ref}})$$

In practice $\text{HR}(x)$ is available either as a smooth function (fractional-polynomial or spline fit to a continuous exposure) or as a step function over k quantile bins, $\text{HR}(x) = \text{HR}_j$ for $x \in \text{bin}_j$. Both forms are supported in the schema.

3. Individual hazard and survival. Under the proportional-hazards assumption, an individual's instantaneous mortality hazard at age $a + t$ is

$$(3) \quad h_i(a + t) = h_0(a + t; s) \cdot \text{HR}(x_i)$$

where h_0 is the age- and sex-specific baseline hazard (from national life tables). The corresponding survival function is

$$(4) \quad S_i(a + t) = \exp\left[-\int_0^t h_i(a + u) du\right] = S_0(a + t) \cdot \text{HR}(x_i)^t$$

4. Propagation to expected years of life lost and DALYs. Comparing an individual's survival curve under their observed value x against the reference curve under x_{ref} gives the biomarker-attributable expected years of life lost:

$$(5) \text{ EYLL}(x; a, s) = \int_0^{\infty} [S_0(a+t) \text{HR}(x_{\text{ref}}) - S_0(a+t) \text{HR}(x)] dt$$

evaluated in practice as a discrete sum over life-table age intervals. Absent a disability-weight adjustment (i.e., treating the biomarker as acting purely through mortality rather than morbidity), the DALY burden attributable to a given biomarker value is taken as

$$(6) \Delta\text{DALY}(x; a, s) = \text{EYLL}(x; a, s)$$

Where morbidity/disability-weight data exist for a biomarker-disease pathway, (6) is extended additively with a YLD (years-lived-with-disability) term; the current dashboard implementation uses the mortality-only form.

5. Expected value of information. We treat the choice of whether to act on a biomarker as a decision problem. Let A be the action space (e.g., {no action, intervene}) and $L(\alpha, x)$ the DALY loss incurred under action $\alpha \in A$ given true biomarker value x . Absent testing, the individual/population selects the action minimizing expected loss under the prior (population) distribution of X :

$$(7) \alpha^* = \operatorname{argmin}_{\alpha \in A} E_{x \sim f_X} [L(\alpha, x)], \quad EL_{\text{no-info}} = E_x [L(\alpha^*, x)]$$

With a test that reveals x (assumed here to be measured without error; measurement error can be folded in as an additional convolution on f_X), the optimal action can be tailored to the realized value:

$$(8) \alpha^*(x) = \operatorname{argmin}_{\alpha \in A} L(\alpha, x), \quad EL_{\text{info}} = E_x [L(\alpha^*(x), x)]$$

The expected value of (perfect) information is then

$$(9) \text{ VOI} = EL_{\text{no-info}} - EL_{\text{info}} \geq 0$$

6. Intervention effectiveness. For a two-action problem {no action, intervene}, with the "no action" loss simply $\Delta\text{DALY}(x)$ from (6), and an intervention assumed capable of shifting the individual's realized value by one standard deviation σ_X of the cohort distribution toward the favorable direction with effectiveness $\varepsilon \in [0,1]$ ($\varepsilon = 1$ denoting full realization of the shift), the intervene-branch loss is

$$(10) L(\text{intervene}, x) = \Delta\text{DALY}(x - \varepsilon \cdot \sigma_X \cdot \text{sign}) + c_{\text{int}}^{\text{DALY}}$$

where sign orients the shift toward lower hazard and $c_{\text{int}}^{\text{DALY}}$ is any DALY-equivalent cost/burden of the intervention itself (e.g., side effects). Combining (6), (9), and (10), the per-individual value of information given realized value x reduces to

$$(11) \text{ VOI}(x) = \max\{ 0, \Delta\text{DALY}(x) - \Delta\text{DALY}(x - \varepsilon \cdot \sigma_{\text{X}} \cdot \text{sign}) - c_{\text{int}} \cdot \text{DALY} \}$$

7. Expected Hazard Information Value (EHIV). To net the DALY-denominated value of a test against its monetary cost, we monetize $\text{VOI}(x)$ using a willingness-to-pay threshold λ per DALY averted (e.g., a national or WHO-referenced cost-effectiveness threshold), and subtract the market price of the test itself, c_{test} (drawn from findalabtest.com list prices and, where available, CMS reimbursement rates for the corresponding CPT/HCPCS code):

$$(12) \text{ EHIV}(a, s, c) = \lambda \cdot E_{\text{X} \sim f_{\text{X}}(x; a, s, c)} [\text{VOI}(x)] - c_{\text{test}}$$

A biomarker test is favorable for a given cohort whenever $\text{EHIV}(a, s, c) > 0$; the dashboard surfaces $\text{EHIV}(a, s, c)$ alongside the underlying $\text{HR}(x)$ and $f_{\text{X}}(x; a, s, c)$ curves for each biomarker so a user can see both the population distribution and the resulting information value at a glance.

8. Cohort dependence. Because f_{X} , HR , σ_{X} , and h_0 are all age-, sex-, and (where data allow) ethnicity- or condition-specific, EHIV is computed per cohort rather than as a single population-wide figure; equation (12) is evaluated separately for each (a, s, c) stratum for which sufficient distributional and hazard data exist. This yields, for a given biomarker, an EHIV surface over age and sex rather than a single scalar.

9. Population/global aggregation. For public-health prioritization, cohort-level VOI is aggregated to a population-level figure by weighting by cohort size $N_{a, s, c}$:

$$(13) \text{ VOI}_{\text{population}} = \sum_{a, s, c} N_{a, s, c} \cdot E_{\text{X} \sim f_{\text{X}}(x; a, s, c)} [\text{VOI}(x)]$$

and, analogously, a population-level EHIV by substituting the aggregate test cost ($N_{a, s, c} \cdot c_{\text{test}}$) into (12). This provides a ranking of biomarkers by total population value, which can be compared against a ranking by per-test EHIV to distinguish tests that are broadly worth deploying at scale from those that are high-value only for narrow, high-risk cohorts.

Notation summary

Symbol	Meaning
x	biomarker value
a, s, c	age, sex, cohort stratifier
$f_{X}(x; a, s, c)$	population density of biomarker X in cohort (a,s,c)
$HR(x)$	mortality hazard ratio at value x, relative to x_{ref}
h_0, S_0	baseline (life-table) hazard and survival
$EYLL(x), \Delta DALY(x)$	biomarker-attributable expected years of life lost / DALY burden
$L(\alpha, x)$	DALY loss under action α given true value x
$VOI, VOI(x)$	expected / realized value of information (DALYs)
σ_X, ε	cohort SD of X; intervention effectiveness
λ	willingness-to-pay threshold per DALY averted
c_{test}, c_{int} DALY	test market price; DALY-equivalent intervention cost
EHIV	Expected Hazard Information Value (monetized net value of testing)
$N_{a,s,c}$	cohort population size

Cohort dependence on VOI (narrative summary)

Value of information exhibits cohort dependence given individual age and sex, as formalized in (12)–(13) above. Additional factors, such as ethnicity, underlying health status, and genetics, can also segment the population into meaningful risk cohorts. We calculate the expected value of information and the associated difference in DALYs assuming the individual acts on the recommendation. Intervention effectiveness ε is another factor shaping VOI; in the initial analysis we assume an individual can shift their marker by one standard deviation ($\varepsilon = 1$ in (10)–(11)), with lower effectiveness values available as a sensitivity parameter.

Global analysis of biomarker VOI

Moving from an individual to a public-health perspective, different biomarkers provide differing value to individuals, and their applicability across a population can vary substantially — captured formally by the population aggregation in (13).

Results

The Physiological Fitness Landscape dashboard

The dashboard itself is available at landscape.opensourcemed.info and allows the user to select a given biomarker. The display shows both the distribution in the general population — with age and sex toggles — and the hazard ratio for mortality (or a proxy) as a function of the biomarker, per equations (1)-(2). The display also reports the expected value of information, EHIV, for each biomarker per equation (12).